Peptic Ulcer Disease (Gastric and Duodenal Ulcers)

All key points for the exam + drug summary table

1. Definition and Causes

- **Peptic ulcer:** a break in the mucosa of the **stomach** (gastric ulcer) or **duodenum** (duodenal ulcer) that extends through the **muscularis mucosae** (usually ≥ 5 mm). Smaller, superficial lesions are **erosions**.
- Results from an imbalance between **aggressive factors** (acid, pepsin, H. pylori, NSAIDs) and **protective factors** (mucus, bicarbonate, **prostaglandins**, mucosal blood flow).

Cause	Key points
Helicobacter pylori	**Most common cause** worldwide (especially **duodenal** ulcers) • Gram-negative, spiral, **urease-producing** bacterium • Also causes gastritis, **gastric cancer**, **MALT lymphoma**
NSAIDs (including low-dose aspirin)	**Second most common** (more often **gastric**) • Inhibit **COX-1** → ↓ protective prostaglandins + direct topical injury • Often **painless** until a complication (bleeding)
Stress-related mucosal damage	Critically ill (ICU) patients
Zollinger-Ellison syndrome (ZES)	**Gastrin-secreting tumor (gastrinoma)** → very high acid → multiple, refractory ulcers
Other	Smoking (impairs healing, ↑ recurrence), cocaine, crack, bisphosphonates, potassium chloride, chemotherapy, Crohn's disease, viral infections (CMV, HSV)

Corticosteroids alone do not clearly cause ulcers, but **combined with NSAIDs** they greatly ↑ ulcer risk. **Anticoagulants, antiplatelets, and SSRIs** ↑ the risk of **bleeding** from an ulcer.

2. Gastric vs Duodenal Ulcer

Feature	Gastric ulcer	Duodenal ulcer
Frequency	Less common	**More common**
Main cause	**NSAIDs** and H. pylori	**H. pylori** (most)
Acid secretion	Normal or low	**Normal or high**
Pain timing	**Worse with food** (soon after eating)	**2–3 hours after meals** and **at night** (wakes patient up)
Effect of food	**Worsens pain** → anorexia, **weight loss**	**Relieves pain** → may gain weight
Malignancy risk	**Yes** → **biopsy** and repeat endoscopy to confirm healing	Rarely malignant
Healing with PPI	**8 weeks**	**4 weeks**

3. Clinical Presentation

Type	Features
Symptoms	**Epigastric pain** (burning, gnawing, "hunger-like") • **Dyspepsia**, bloating, nausea, early satiety, belching • Pain may be relieved by **antacids** • **NSAID ulcers are often silent**
Alarm features	**GI bleeding** (**hematemesis**, **"coffee-ground" vomit**, **melena** = black tarry stools) • **Iron deficiency anemia** • **Unintentional weight loss** • **Dysphagia** • **Persistent vomiting** • Palpable mass • Family history of GI cancer • **New dyspepsia at age ≥ 60**
Perforation	**Sudden, severe abdominal pain**, rigid abdomen → **surgical emergency**

Approach to dyspepsia (ACG)

- **Age ≥ 60 → upper endoscopy**.
- **Age < 60 → test for H. pylori** (non-invasive) and **treat if positive** ("**test and treat**"); if negative or symptoms persist → **empiric PPI**.
- Alarm features at any age → consider endoscopy.

4. Complications

Complication	Key points
Upper GI bleeding	**Most common** complication • Hematemesis, melena, anemia, hypotension
Perforation	Sudden severe pain, peritonitis • Surgery • More common with NSAIDs
Gastric outlet obstruction	Early satiety, vomiting, weight loss (scarring near pylorus)
Penetration	Into adjacent organs (e.g., pancreas → back pain)

5. Diagnosis

Endoscopy

- **Gold standard** for diagnosing ulcers • Allows **biopsy** (gastric ulcers to rule out cancer; H. pylori testing) and **treatment of bleeding**.

H. pylori tests

Test	Type	Key points
Urea breath test (UBT)	Non-invasive	Detects **active infection** • **Preferred** for diagnosis and **confirming eradication**
Stool antigen test	Non-invasive	Detects **active infection** • Diagnosis and **confirming eradication**
Serology (antibodies)	Non-invasive	Stays positive after cure → **cannot distinguish active vs past infection** • **Not recommended**; **never** use to confirm eradication
Rapid urease test (CLO test)	Invasive (biopsy)	Fast result during endoscopy
Histology	Invasive (biopsy)	Also shows gastritis, metaplasia
Culture + susceptibility (or molecular testing)	Invasive (biopsy) or stool PCR	Guides therapy (**clarithromycin, levofloxacin** resistance) • Recommended after treatment failure

Avoid false-negative results: stop **PPIs for 2 weeks** and **antibiotics and bismuth for 4 weeks** before UBT, stool antigen, or biopsy tests. **H2RAs** and antacids can be used meanwhile.

Who should be tested for H. pylori

- **Active or past peptic ulcer** (unless cure was documented)
- **Dyspepsia** in patients < 60 years (test and treat)
- **Gastric MALT lymphoma**, early gastric cancer, family history of gastric cancer
- Starting **long-term NSAIDs** or on **long-term low-dose aspirin**
- **Unexplained iron deficiency anemia**, **idiopathic thrombocytopenic purpura (ITP)**
- Household members of a patient with H. pylori (current guidance)

6. H. pylori Eradication Therapy (ACG 2024)

General principles

- **Treat everyone who tests positive.**
- **Duration: 14 days** for all regimens.
- **PPI must be given twice daily** (high dose) to raise gastric pH and improve antibiotic activity.
- **Clarithromycin-based regimens** only if the strain is **known to be susceptible** (resistance is common; avoid if **prior macrolide use** or local resistance > 15%).
- **Adherence** is the most important patient factor for success.
- **Always confirm eradication** (see below).

First-line regimens (treatment-naïve)

Regimen	Drugs and doses (14 days)	Notes
Bismuth quadruple therapy (BQT) — preferred	**PPI BID** + **bismuth subcitrate 120–300 mg or subsalicylate 300 mg QID** + **tetracycline 500 mg QID** + **metronidazole 500 mg TID–QID**	**Preferred** when susceptibility is unknown • **Penicillin allergy:** regimen of choice • Pylera capsules (bismuth + metronidazole + tetracycline) 3 caps QID + omeprazole 20 mg BID
Rifabutin triple therapy (Talicia)	**Omeprazole 40 mg + amoxicillin 1 g + rifabutin 50 mg TID** (= **4 capsules TID with food**)	Alternative if no penicillin allergy • Rifabutin is a **CYP3A4 inducer**
Vonoprazan dual therapy	**Vonoprazan 20 mg BID** + **amoxicillin 1 g TID**	Alternative if no penicillin allergy
Clarithromycin triple therapy	**PPI BID** + **clarithromycin 500 mg BID** + **amoxicillin 1 g BID** (or **metronidazole 500 mg BID–TID** if penicillin-allergic)	**Only if clarithromycin susceptibility is confirmed**
Vonoprazan triple therapy	**Vonoprazan 20 mg BID** + **clarithromycin 500 mg BID** + **amoxicillin 1 g BID**	Only if clarithromycin susceptibility is confirmed or unknown with no better option

Older regimens (may still appear on exams)

Regimen	Drugs (10–14 days)
Concomitant therapy	PPI BID + clarithromycin 500 mg BID + amoxicillin 1 g BID + metronidazole/tinidazole 500 mg BID
Sequential therapy	Days 1–7: PPI + amoxicillin • Days 8–14: PPI + clarithromycin + metronidazole/tinidazole
Hybrid therapy	PPI + amoxicillin for 7 days, then PPI + amoxicillin + clarithromycin + metronidazole for 7 days

Salvage (after treatment failure)

- **Do not repeat clarithromycin or levofloxacin** if used before (resistance) • Ideally use **susceptibility testing**.
- Failed a **non-bismuth** regimen → **bismuth quadruple therapy**.
- Failed **bismuth quadruple** → **rifabutin triple therapy**, or **levofloxacin triple** if susceptible.
- **Levofloxacin triple:** PPI BID + **levofloxacin 500 mg daily** + **amoxicillin 1 g BID** × 14 days.
- **Penicillin allergy:** consider **allergy evaluation (delabeling)** — opens more options.

Confirming eradication

- Test **at least 4 weeks after finishing** antibiotics, with **PPI stopped for 2 weeks**.
- Use **urea breath test**, **stool antigen**, or biopsy — **never serology**.

Key adverse effects and interactions of H. pylori drugs

Drug	Counseling points
Bismuth (subsalicylate, subcitrate)	**Black tongue and black stools** (harmless) • Subsalicylate: **salicylate** → avoid with **aspirin allergy**, caution with **warfarin**, children/teens with viral illness (**Reye syndrome**), renal impairment
Metronidazole	**Metallic taste**, nausea • **Disulfiram-like reaction** → **no alcohol during and for 3 days after** • ↑ **Warfarin effect (INR)** • Peripheral neuropathy (long use) • ↑ lithium
Tetracycline	**Avoid in pregnancy and children < 8** (teeth discoloration, bone) • **Photosensitivity** • **Pill esophagitis** (full glass of water, stay upright) • **Chelation** with calcium, iron, antacids, dairy
Clarithromycin	**Strong CYP3A4 inhibitor** → ↑ **simvastatin, lovastatin** (contraindicated), **colchicine**, **calcium channel blockers**, **warfarin**, carbamazepine, tacrolimus • **QT prolongation** • Metallic taste, GI upset
Amoxicillin	**Rash**, **diarrhea** • Avoid in **penicillin allergy**
Levofloxacin	**Tendinopathy/rupture**, **QT**, CNS effects, **dysglycemia**, aortic aneurysm • **Chelation** with antacids, bismuth, iron, calcium (separate)
Rifabutin	**Orange-red body fluids** (urine, tears; stains contact lenses) • **CYP3A4 inducer** → ↓ **hormonal contraceptives**, many drugs • Neutropenia, uveitis
Vonoprazan	CYP3A4 substrate • **Contraindicated with rilpivirine** • Diarrhea
All antibiotics	Diarrhea, **C. difficile** risk • **Finish the full 14-day course**

7. NSAID-Induced Ulcers

Risk factors for NSAID GI complications

- **History of ulcer or GI bleeding** (**strongest risk factor**, especially a complicated ulcer)
- **Age > 65** (some references > 60)
- **High-dose NSAIDs** or **more than one NSAID** (including **low-dose aspirin**)
- Concurrent **anticoagulants**, **antiplatelets**, **corticosteroids**, **SSRIs/SNRIs**
- **H. pylori infection**
- Serious comorbidity

Risk stratification and prevention (ACG)

GI risk	Low CV risk	High CV risk (needs low-dose aspirin)
Low (no risk factors)	NSAID alone (lowest effective dose)	**Naproxen + PPI**
Moderate (1–2 risk factors)	**NSAID + PPI** (or misoprostol)	**Naproxen + PPI**
High (> 2 risk factors, prior complicated ulcer, or on anticoagulants/steroids)	**Alternative therapy**, or **COX-2 inhibitor (celecoxib) + PPI**	**Avoid NSAIDs** — use alternative therapy

Prevention options

Option	Key points
PPI (standard dose daily)	**Preferred** • Prevents gastric and duodenal ulcers
Misoprostol (prostaglandin E1 analog)	**200 mcg QID** (100 mcg if not tolerated) • As effective as PPI but **poorly tolerated**: **diarrhea**, abdominal cramps • **Boxed warning: contraindicated in pregnancy** (**abortion**, birth defects) • Women of childbearing age need negative pregnancy test and contraception
H2RA (double dose, e.g., famotidine 40 mg BID)	**Less effective** than PPI; standard doses do **not** prevent gastric ulcers
COX-2 selective inhibitor (celecoxib)	**Lower GI risk** than nonselective NSAIDs • **↑ CV risk** (boxed warning) • Benefit is lost if taken with aspirin • High-risk patients: celecoxib + PPI
Combination products	Diclofenac/misoprostol (Arthrotec) • Naproxen/esomeprazole (Vimovo) • Ibuprofen/famotidine (Duexis)
Test and treat H. pylori	Before starting long-term NSAIDs
Sucralfate	**Not effective** for NSAID ulcer prevention

Treating an NSAID-induced ulcer

- **Stop the NSAID** if possible.
- **PPI** for **8 weeks** (gastric ulcer) or **4 weeks** (duodenal ulcer); **longer** if the NSAID must be continued.
- If the NSAID must continue → **continue PPI**, use the lowest dose, consider COX-2 inhibitor.
- **Test for H. pylori** and treat if positive.

Low-dose aspirin and antiplatelet users

- Aspirin for **secondary CV prevention** → **continue or resume quickly** after bleeding (as soon as hemostasis is achieved, usually within a few days) **+ PPI**; stopping aspirin ↑ CV events and death.
- **Aspirin for primary prevention** → usually **stop** after a GI bleed.
- **Clopidogrel + PPI** → use **pantoprazole** (avoid omeprazole/esomeprazole: CYP2C19 inhibition).
- **Dual antiplatelet therapy (DAPT)** or anticoagulant + antiplatelet → **PPI** recommended for GI protection.

8. Ulcer Healing Therapy

Drug	Dose	Duration
PPI (preferred)	Standard dose once daily (omeprazole 20 mg, esomeprazole 40 mg, lansoprazole 30 mg, pantoprazole 40 mg, rabeprazole 20 mg, dexlansoprazole 60 mg)	**Duodenal: 4 weeks • Gastric: 8 weeks** • Longer if large/complicated ulcer, NSAID continued, or refractory
H2RA	e.g., **famotidine 40 mg at bedtime** or 20 mg BID • cimetidine 800 mg HS • nizatidine 300 mg HS	8 weeks
Sucralfate	**1 g QID** (or 2 g BID) on an empty stomach	4–8 weeks (duodenal ulcer)

- **H. pylori-positive ulcer:** the **eradication regimen** (with its BID PPI) is usually enough for small, uncomplicated duodenal ulcers; **continue PPI after eradication** for large or complicated ulcers and gastric ulcers.
- **Maintenance PPI** (long term) for: recurrent ulcers, H. pylori-negative ulcers, ongoing NSAID/aspirin need, failed eradication, ZES.
- **Gastric ulcer** → **repeat endoscopy** (8–12 weeks) to confirm healing and exclude cancer.

9. Upper GI Bleeding from Ulcers

1. **Resuscitation:** IV fluids • **Transfuse if Hgb < 7 g/dL** (**< 8** with cardiovascular disease).
2. **Risk assessment: Glasgow-Blatchford score** (score 0–1 → may be managed as outpatient).
3. **Pre-endoscopy:** IV PPI may be given (reduces high-risk findings; does not change outcomes) • **Erythromycin 250 mg IV** 30–120 min before endoscopy (prokinetic: **improves visualization**).
4. **Endoscopy within 24 hours** → endoscopic hemostasis for high-risk stigmata (active bleeding, visible vessel, adherent clot).
5. **After endoscopic hemostasis** (high-risk ulcer): **high-dose PPI for 72 hours — 80 mg IV bolus** then **8 mg/h continuous infusion**, **or** intermittent high-dose IV PPI (e.g., 40–80 mg IV BID) as an alternative.
6. Then **oral PPI twice daily** through day 14, then **once daily** (duration based on ulcer cause).
7. **Test for H. pylori** (if negative during acute bleeding, **repeat the test later** — false negatives are common) and treat if positive.
8. **Stop NSAIDs**; resume **aspirin for secondary prevention** as soon as possible with PPI.

Not recommended for ulcer bleeding: tranexamic acid (no benefit, ↑ VTE) • **Octreotide** is for **variceal** bleeding, not ulcer bleeding.

10. Stress Ulcer Prophylaxis (SUP)

Point	Details
Who	**ICU patients** with risk factors: **mechanical ventilation > 48 hours** • **Coagulopathy** (platelets < 50,000, INR > 1.5) • **History of GI bleeding** within 1 year • Shock/sepsis • Traumatic brain or spinal cord injury • Severe burns • High-dose corticosteroids • Acute kidney or liver failure
Who NOT	**Non-ICU (general ward) patients** without risk factors — **do not give SUP**
Agents	**PPI** (e.g., pantoprazole 40 mg IV/PO daily) — **more effective** in preventing bleeding • **H2RA** (e.g., **famotidine 20 mg IV/PO BID**) — alternative • Sucralfate (less effective; aluminum, drug binding, tube clogging)
Risks	**Pneumonia** (ventilator-associated), **C. difficile infection**
Stop	**When risk factors resolve** or at ICU transfer • **Do not continue at hospital discharge** (common error)

11. Zollinger-Ellison Syndrome (ZES)

- **Gastrinoma** (pancreas or duodenum) → ↑ **gastrin** → very high acid secretion; ~25% are part of **MEN1**.
- Features: **multiple**, **refractory**, or unusually located ulcers (e.g., jejunum) • **Diarrhea** • GERD.
- Diagnosis: **high fasting gastrin** with low gastric pH (PPIs raise gastrin → interpret carefully).
- **Treatment: high-dose PPI** (e.g., omeprazole **60 mg daily**, titrated up to **120 mg TID**) • **Surgical removal** of the tumor • Octreotide for some cases.

12. Refractory or Recurrent Ulcers

Ulcer that does not heal after **8–12 weeks** of PPI. Look for:

1. **Nonadherence**
2. **Persistent H. pylori** (failed eradication — retest)
3. **Continued NSAID or aspirin use** (including **hidden OTC NSAIDs** and combination products)
4. **Smoking**
5. **Zollinger-Ellison syndrome** (hypersecretion)
6. **Malignancy** (gastric ulcer)
7. Other causes: Crohn's disease, infections (CMV), ischemia, cocaine

13. Lifestyle and Counseling

- **Stop smoking** (delays healing, ↑ recurrence).
- **Avoid NSAIDs** (including OTC ibuprofen, naproxen, aspirin, and combination cold products) → use **acetaminophen** for pain.
- **Limit alcohol**.
- Avoid foods/drinks that worsen symptoms (no specific diet heals ulcers; **spicy food does not cause ulcers**).
- Reduce stress.
- Recognize **bleeding signs**: black stools, vomiting blood or coffee grounds, dizziness → seek care.
- **Complete the full H. pylori course** and return for the eradication test.

14. Special Populations

Population	Key points
Pregnancy	**Avoid**: **misoprostol** (abortifacient), **tetracycline**, **bismuth subsalicylate**, clarithromycin (caution), levofloxacin • **Defer H. pylori eradication** until after delivery when possible • Antacids, sucralfate, H2RA, PPI acceptable
Children	**No tetracycline < 8 years** • Avoid bismuth subsalicylate (Reye) and fluoroquinolones • Weight-based dosing
Elderly	Higher NSAID ulcer and bleeding risk • New dyspepsia ≥ 60 → endoscopy • Avoid cimetidine • PPI long-term risks
Renal impairment	Adjust **H2RAs**, **tetracycline** (avoid), clarithromycin (CrCl < 30) • Avoid magnesium/aluminum antacids and bismuth long-term • PPIs: no adjustment

Exam Traps and Frequently Tested Points

1. **Most common causes: H. pylori** and **NSAIDs**.
2. **Duodenal ulcer:** pain **2–3 h after meals and at night**, **relieved by food**, mostly H. pylori.
3. **Gastric ulcer:** pain **worse with food**, weight loss, **biopsy** (malignancy risk), more NSAIDs.
4. **Healing:** duodenal **4 weeks**, gastric **8 weeks** of PPI.
5. **Most common complication: bleeding**.
6. **NSAID ulcers** are often **painless** until they bleed.
7. **Dyspepsia ≥ 60 → endoscopy**; < 60 → **test and treat** H. pylori.
8. **UBT and stool antigen** = active infection; **serology cannot confirm cure**.
9. **Before testing:** stop **PPIs 2 weeks**, **antibiotics and bismuth 4 weeks**.
10. **Confirm eradication ≥ 4 weeks** after therapy (off PPI 2 weeks).
11. **All H. pylori regimens: 14 days**, **PPI twice daily**.
12. **Bismuth quadruple** (PPI + bismuth + tetracycline + metronidazole) = **preferred first-line** and for **penicillin allergy**.
13. **Clarithromycin triple** only if **clarithromycin susceptibility is known**; avoid after previous macrolide use.
14. **New options: rifabutin triple** (Talicia) and **vonoprazan dual** (vonoprazan + amoxicillin).
15. **Salvage:** don't repeat clarithromycin or levofloxacin • failed BQT → rifabutin or levofloxacin triple.
16. **Bismuth → black tongue/stools** • subsalicylate → avoid in aspirin allergy, warfarin, children (Reye).
17. **Metronidazole → no alcohol** (disulfiram-like) during and **3 days after**; ↑ INR with warfarin; metallic taste.
18. **Tetracycline** → not in pregnancy or < 8 years; photosensitivity; chelation.
19. **Clarithromycin → CYP3A4 inhibitor** (simvastatin, lovastatin, colchicine) + **QT**.
20. **Rifabutin** → orange body fluids, ↓ **oral contraceptives**.
21. **Strongest NSAID ulcer risk factor: previous ulcer/bleed**.

22. **NSAID prevention: PPI preferred**; **misoprostol** effective but diarrhea + **contraindicated in pregnancy**; standard-dose H2RA and sucralfate **do not** prevent NSAID gastric ulcers.
23. **High GI + high CV risk → avoid NSAIDs** • High CV risk needing NSAID → **naproxen + PPI**.
24. **Celecoxib** → less GI risk but ↑ CV risk; GI benefit lost with aspirin.
25. **Clopidogrel + PPI → pantoprazole** (not omeprazole/esomeprazole).
26. **Aspirin for secondary prevention → resume early** after bleeding + PPI.
27. **Ulcer bleeding after endoscopy: PPI 80 mg IV bolus + 8 mg/h × 72 h** (or intermittent high dose), then PO BID to day 14.
28. **Erythromycin IV** before endoscopy improves visualization • **Tranexamic acid** not recommended • **Octreotide** = varices.
29. **Transfuse at Hgb < 7** (< 8 with CVD).
30. **SUP** only for **ICU patients with risk factors** (ventilation > 48 h, coagulopathy) • **stop at discharge**.
31. **ZES** → multiple refractory ulcers + diarrhea + high gastrin → **high-dose PPI**.
32. **Refractory ulcer** → check adherence, H. pylori, hidden NSAIDs, smoking, ZES, cancer.
33. **Smoking** delays ulcer healing • **acetaminophen** is the preferred analgesic.

Note: Some doses and recommendations differ between references and guideline versions (e.g., older guidelines allowed clarithromycin triple therapy first-line). **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Omeprazole	Healing 20 mg daily • H. pylori 20 mg BID • ZES 60 mg daily (up to 120 mg TID)	Headache, diarrhea • long-term: **C. diff, fractures, ↓Mg, ↓B12**, pneumonia, AIN	**CYP2C19 inhibitor** → ↓ **clopidogrel** (avoid) • ↑ citalopram, diazepam, phenytoin, MTX • ↓ azoles, atazanavir, **rilpivirine (CI)**
Esomeprazole	Healing 20–40 mg daily • H. pylori 40 mg BID • IV available	Same as PPIs	Same as omeprazole (**avoid with clopidogrel**)
Lansoprazole	Healing 15–30 mg daily • H. pylori 30 mg BID	Same as PPIs	pH-dependent DDIs
Pantoprazole	Healing 40 mg daily • H. pylori 40 mg BID • GI bleed: 80 mg IV bolus + 8 mg/h × 72 h • SUP 40 mg daily	Same as PPIs	**Preferred with clopidogrel** • fewest CYP DDIs
Rabeprazole	Healing 20 mg daily • H. pylori 20 mg BID	Same as PPIs	Less CYP2C19 dependence
Dexlansoprazole	30–60 mg daily	Same as PPIs	No regard to meals
Vonoprazan	H. pylori: 20 mg BID (dual or triple) × 14 days	Diarrhea, abdominal pain	**CYP3A4 substrate • rilpivirine CI** • with or without food
Famotidine	Healing 40 mg HS or 20 mg BID • SUP 20 mg IV/PO BID • NSAID prevention 40 mg BID	Headache • **confusion** (elderly/renal) • QT (renal)	**Reduce if CrCl < 60** • no CYP inhibition • pH-dependent DDIs
Cimetidine	800 mg HS or 400 mg BID	**Gynecomastia**, impotence, confusion	**CYP inhibitor** → ↑ **warfarin, theophylline, phenytoin**, lidocaine
Sucralfate	1 g QID (or 2 g BID) on empty stomach	**Constipation**, aluminum accumulation (CKD)	**Binds** quinolones, levothyroxine, phenytoin, warfarin, digoxin → **separate 2 h** • not for NSAID ulcer prevention

Drug	Dose	Key ADRs	Key DDIs / special notes
Misoprostol	200 mcg QID with food (100 mcg if not tolerated)	**Diarrhea**, abdominal cramps	**Boxed: contraindicated in pregnancy** (abortion, birth defects) • avoid Mg antacids (worse diarrhea)
Bismuth subsalicylate	300 mg (2 × 262 mg tabs ≈ 524 mg) QID	**Black tongue/stools**, constipation • salicylate toxicity (tinnitus)	Avoid in **aspirin allergy**, with **warfarin**, children/teens (**Reye**) • chelates tetracyclines/quinolones (separate when possible)
Bismuth subcitrate potassium	120–300 mg QID (Pylera: 3 caps QID)	Black tongue/stools	No salicylate
Tetracycline	500 mg QID × 14 days	GI upset, **photosensitivity**, **esophagitis** • tooth discoloration	**CI in pregnancy, children < 8** • chelation with Ca, Fe, Mg, Al, dairy • avoid in renal impairment
Metronidazole	500 mg TID–QID (BQT) • 500 mg BID (triple)	**Metallic taste**, nausea, **peripheral neuropathy**, dark urine	**Disulfiram-like reaction** (no alcohol during + 3 days after) • ↑ **warfarin** INR • ↑ lithium • avoid with disulfiram
Clarithromycin	500 mg BID × 14 days	Metallic taste, GI upset, **QT prolongation**	**Strong CYP3A4 inhibitor** → ↑ **simvastatin, lovastatin (CI)**, **colchicine**, CCBs, warfarin, carbamazepine, tacrolimus • avoid with other QT drugs
Amoxicillin	1 g BID (triple) • 1 g TID (vonoprazan dual)	Rash, diarrhea	Avoid in penicillin allergy • ↑ rash with allopurinol
Levofloxacin	500 mg daily × 14 days (salvage)	**Tendon rupture**, **QT**, CNS, dysglycemia, aortic aneurysm	**Chelation** with antacids, bismuth, iron, Ca (take 2 h before / 6 h after) • ↑ warfarin • only if susceptible
Rifabutin (Talicia: omeprazole 40 mg + amoxicillin 1 g + rifabutin 50 mg)	4 capsules TID with food × 14 days	**Orange-red body fluids**, neutropenia, uveitis	**CYP3A4 inducer** → ↓ **oral contraceptives** (use backup), ↓ many drugs
Erythromycin (pre-endoscopy)	250 mg IV 30–120 min before endoscopy	QT prolongation, GI cramps	CYP3A4 inhibitor
Celecoxib	100–200 mg BID (or 200 mg daily)	**Boxed: CV events, GI bleeding** • renal toxicity, HTN	**Sulfonamide allergy** caution • GI benefit lost with aspirin • CYP2C9 substrate (fluconazole ↑)

Good luck on your exam!